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Original article

Safe reinforcement learning based agile satellite attitude control for non-cooperative target tracking with operation constraints and actuator faults

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ABSTRACT

Persistent tracking of space-based non-cooperative targets is a fundamental capability for achieving space situational awareness and enabling numerous downstream space applications. However, it presents significant challenges, particularly during agile attitude maneuvers, due to the need to respect operational constraints and handle potential actuator faults. In this work, we divide the agile satellite tracking task into three distinct stages and formulate it as a constrained spacecraft attitude control problem. To systematically address the complex practical challenges in agile satellite control–including attitude constraints from forbidden zones, angular velocity limitations, torque saturation, and potential actuator faults–we formulate the problem as a Constrained Markov Decision Process (CMDP). To enhance the adaptability and fault-tolerance of the controller, we integrate a data-driven fault identification module capable of estimating both actuator effectiveness loss and additive torque faults. A safe reinforcement learning framework is further employed to learn fault-tolerant control policies that ensure continuous and stable tracking performance while minimizing the risk of constraint violations. Extensive simulation results demonstrate that the proposed approach enables the agile imaging satellite to achieve accurate, persistent tracking under strict operational constraints, while maintaining safe and resilient attitude control in the presence of actuator anomalies.

1. Introduction

In non-cooperative target tracking missions, the implementation of active tracking strategies holds irreplaceable significance for achieving space situational awareness [1,2] and enabling numerous downstream space applications. Passive, wait-and-see observation methods are wholly inadequate for meeting mission requirements, as targets may exhibit unexpected motions necessitating real-time response capabilities from the tracking system. These core requirements render active tracking the sole viable solution for addressing non-cooperative target observation challenges. Active tracking of space non-cooperative targets is a crucial prerequisite for downstream tasks such as attitude estimation [3] and pose tracking [4]. The tracking of non-cooperative space targets presents unique challenges due to their orbital characteristics. With severely limited visibility windows per orbit, these targets may abruptly migrate from detectable ranges to unobserved regions. Unlike conventional Earth-pointing or wait-and-see observation missions in aerial target tracking [5], tracking non-cooperative targets in space requires agile satellites to perform rapid responded large-angle attitude maneuvers to maintain persistent line-of-sight alignment. These agile reorientations present significant operational challenges, particularly in protecting sensitive onboard payloads.

During high-speed attitude maneuvers, inadvertent exposure to bright celestial bodies may cause irreversible damage or permanent degradation to optical instruments and imaging sensors. To ensure payload safety, the satellite must avoid predefined exclusion zones in the attitude space–known as forbidden-pointing regions. These constraints are typically time-varying, requiring careful designation to ensure continuous safety satisfaction throughout the mission. Moreover, to preserve actuator health and ensure feasible control actions, additional dynamic constraints such as angular velocity limits and torque saturation bounds must be considered. These constraints are often coupled and jointly restrict the maneuverability of the satellite, especially during aggressive tracking. In addition to operational constraints, another critical

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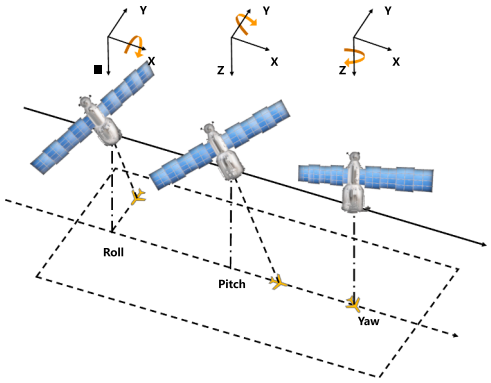


Fig. 1. The attitude Maneuver capability of the Agile Satellite.

challenge arises from actuator faults, which are frequently encountered during long-duration autonomous missions in space. Actuator faults can manifest as effectiveness degradation (modeled as multiplicative attenuation) or additive biases in the output torque. If not timely identified and compensated, such faults may degrade tracking accuracy or lead to mission failure. However, most existing studies on spacecraft tracking and control either ignore faults or rely on ideal assumptions, limiting their applicability to realistic fault-prone environments.

To address the high-velocity characteristics of non-cooperative targets, our approach leverages the exceptional attitude control ability of agile satellites (as demonstrated in Fig. 1): (1) During the initial acquisition phase, the satellite executes rapid large-angle maneuvers to promptly align its payload field-of-view with the target; (2) In the subsequent observation/tracking stage, it switches to high-precision attitude fine-tuning mode to optimize tracking accuracy and observation quality. Furthermore, to address the complex challenges of non-cooperative space target tracking, we formulate the active tracking problem as a constrained attitude control problem under uncertainty. Our approach explicitly considers both operational constraints and potential actuator faults. The problem is formally cast as a Constrained Markov Decision Process (CMDP) [6], which provides a unified framework for: (i) simultaneous handling of multiple operational constraints, (ii) robust performance under actuator fault scenarios, and (iii) optimization of long-term tracking objectives while maintaining safety guarantees.

To this end, we propose a fault-tolerant attitude control framework that integrates safe reinforcement learning (SRL) [7] with data-driven fault identification, offering a robust solution for real-world tracking missions in complex space environments. At the technical implementation level, our approach innovatively integrates spatial satellite-target position vector information with attitude into a unified representation of error quaternions. Concurrently, we innovatively incorporate payload field-of-view angle into the reinforcement learning reward function design, enhancing the tracking process whilst ensuring the target remains continuously within the field of view. Through careful formulation, our method ensures reliable tracking performance while minimizing the risk of constraint violations during critical operations.

Our main contributions are summarized as follows:

- We introduce, to our knowledge, the first Constrained Markov Decision Process framework that systematically addresses the complete spacecraft attitude control challenge for active non-cooperative target tracking. The complex tracking task is divided into three sequential, interdependent stages, which incorporate critical real-world constraints: (i) forbidden-pointing regions, (ii) angular velocity limits, (iii) torque saturation, and (iv) actuator faults. The proposed formulation establishes safety guarantees throughout all mission phases while maintaining tracking performance.
- We develop a neural network-based time-series regression method for spacecraft actuator fault identification, which enables efficient and accurate estimation of both effectiveness loss and additive faults during on-orbit operations.
- We adopt a safe off-policy reinforcement learning algorithm–WCSAC (Worst-case Soft Actor-Critic)–to train fault-tolerant control policies. Under the specialized reward/penalty structure that accounts for mission-specific constraints, the learned policy ensures reliable attitude tracking performance under varying operational constraints and fault scenarios while minimizing the risk of constraint violations.

2. Related work

2.1. Active tracking of non-cooperative target

Active tracking of non-cooperative targets for observation satellites involves a large-angle pre-maneuver phase to align the field of view with the target and a fine-angle post-adjustment phase to optimize observation accuracy.

Most existing research on active tracking primarily emphasizes computer vision techniques during the fine-angle post-adjustment phase [8,9], where high-precision tracking is achieved through visual feedback and fine-grained control. However, limited attention has been given to the control mechanisms during the large-angle pre-maneuver phase, particularly in terms of attitude dynamics and payload pointing control. The challenge lies in accurately following the target's motion via the rapid and accurate attitude maneuvers of the imaging satellite. Recent studies have made significant progress in this area by leveraging advanced control strategies. Cao et al. [5] addressed the dynamic process of target tracking by modeling it as a Markov state transition, which takes into consideration the movement of the moving target and the prediction model of target position. Wu et al. [10] proposed a novel attitude tracking strategy for hybrid actuators, dividing the tracking mission into three distinct processes and validating strategy accuracy through numerical simulations. Building on this work, Wu et al. [11] achieved high-precision attitude tracking by employing an improved back-stepping controller. To tackle the uncertainties inherent in tracking operations, Zheng et al. [12] developed a reinforcement learning-based sliding mode observer, which effectively estimates and compensates for these uncertainties. Yu et al. [9] proposed a reinforcement learning-based action decision strategy for chaser satellites, leveraging extracted visual object information. Taking safe motion constraints into account, Lin et al. [13] combined the back-stepping control method and an adaptive sliding mode law to achieve robust 6 DOF tracking. Lu et al. [14] utilized reinforcement learning to govern satellite attitude control through torque outputs, achieving both accurate and stable tracking of moving targets. Additionally, Lu et al. [15] proposed a satellite constellation scheduling method for multi-target tracking, extending their reinforcement learning-based attitude controller to complex multi-object scenarios.

While studies such as [8,9] have attempted to integrate visual object tracking with chaser spacecraft control, their methods simplify the action space of chaser spacecraft to coordinate displacements, neglecting the detailed consideration of command torques and attitude dynamics. Moreover, practical considerations such as operational constraints [16–20] and actuator faults [21] are critical for real-world applications and require further investigation. In summary, while significant progress has been made in active tracking of non-cooperative targets, there remains a need for more comprehensive approaches that integrate advanced control strategies with practical operation considerations to enhance the robustness and applicability of these systems in real-world scenarios.

2.2. Reinforcement learning in spacecraft control

Recent advances in reinforcement learning [22–27] have demonstrated promising potential for spacecraft attitude control applications. Traditional control approaches have shown limitations in

handling complex operation constraints [16–19] and unexpected nonlinear disturbances or faults [21], as well as the control torque saturation problem [28,29]. In contrast, RL-based methods offer several advantages, including adaptability to uncertain environments and the ability to learn optimal control policies through interaction with the environment. In addressing rapid attitude transitions during complex orbital maneuvers, Ma et al. [30] developed a deep reinforcement learning approach that achieved significantly better stabilization performance than conventional proportional-derivative control methods. Elkins et al. [22,23] proposed an innovative adaptive control architecture using deep reinforcement learning, specifically designed for large-angle reorientation tasks while maintaining precise pointing accuracy. The study not only validated the controller's ability to meet stringent industry accuracy requirements but also introduced methodological improvements through reward function optimization. Liu et al. [31] proposed a reinforcement learning-based attitude tracking control strategy for combined spacecraft takeover maneuvers subject to completely unknown dynamics. Lu et al. [32] introduced a RL-based fault-tolerant control framework for the spacecraft attitude control problem, incorporating carefully designed state representations and a hybrid reward mechanism. The proposed framework ensures both rapid response and stable fault recovery, improving overall control robustness. Recent advances in safe reinforcement learning [33] have shown significant promise for addressing critical spacecraft control challenges, which provide a principled framework for developing controllers that can satisfy stringent safety requirements while maintaining operational effectiveness. Mu et al. [34] formulated the spacecraft collision avoidance maneuver problem as a constrained Markov decision process, incorporating safety constraints directly into the learning process. This approach achieved autonomous collision avoidance in simulation scenarios, addressing a key challenge in applying RL to safety-critical systems. Taking contingent on obstacle warnings and collision avoidance constraints into account, Sharma et al. [35] established a safety reinforcement learning framework for devising spacecraft rendezvous guidance strategies.

In the present study, we extend the safe reinforcement learning methodology to the domain of non-cooperative target tracking. Our formulation treats the space target tracking problem as a specialized instance of attitude control, where the agent must continuously maintain the target within the payload's field of view while respecting operational constraints.

3. Preliminary

3.1. Task decomposition

The task decomposition for space target tracking is depicted in Fig. 2, with its operational workflow analyzed in the context of an agile imaging satellite mission. The entire process consists of three critical stages: Fault Identification Stage (A-B), Attitude Adjustment Stage (B-C), and Target Tracking Stage (C-D).

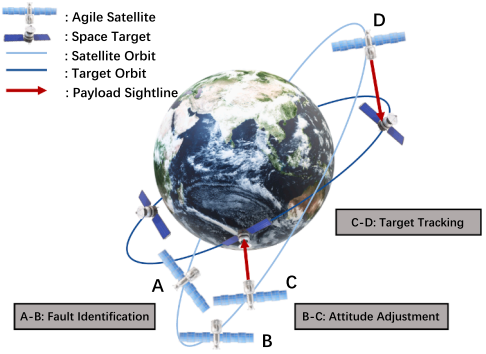


Fig. 2. Task decomposition.

3.1.1. Fault identification stage

The fault identification process follows a systematic approach similar to traditional spacecraft parameter identification, which utilizes the time series of command torques and measured attitude responses (such as angular velocity and quaternion) to estimate actuator faults through model-based parameter estimation. By analyzing the discrepancy between expected and observed dynamics under known torque commands, both effectiveness loss and additive bias can be determined using optimization techniques or neural networks.

3.1.2. Attitude adjustment stage

Following successful fault identification, the satellite must rapidly reorient itself from an arbitrary initial state to align its payload's sightline with the space target. This stage accounts for the identified actuator faults to ensure reliable maneuver execution. At the beginning of this stage, the satellite maintains an arbitrary attitude and angular velocity configuration. Upon command reception, the satellite must initiate precise attitude maneuvers to ensure target acquisition within the limited orbital visible time window. The satellite's maneuverability is demonstrated through its three-axis (roll, pitch, and yaw) control capability [14]. This advanced maneuverability enables the satellite to perform large-angle attitude adjustments, facilitating target acquisition within the payload's field-of-view [36].

During the attitude adjustment stage, our primary objective is to align the payload's sightline with the relative position vector from the agile satellite to the space target. Following the established methodology in [14,37], we adopt the convention that the payload is mounted along the satellite body's z-axis. Denote $\overrightarrow{OT}_{ECI}$ and $\overrightarrow{OS}_{ECI}$ represent the ECI position vector of the space target and the agile satellite, respectively. The alignment of ST_{ECI} and payload sightline z_{body} can be mathematically expressed as:

$$z_b = \frac{\overrightarrow{ST}_{ECI}}{\left|\overrightarrow{ST}_{ECI}\right|} \tag{1}$$

To ensure the staring attitude where there is no rotation of the z-axis, the x-axis x_b and y-axis y_b vectors in the body frame can be obtained via:

$$\begin{cases} x_b = \dfrac{z_b \times (-y_o)}{|z_b \times (-y_o)|} \\ y_b = z_b \times x_b \end{cases} \tag{2}$$

The desired Euler angles under the yaw-pitch-roll (3-2-1) rotation sequence can be derived by equating the current rotation matrix with the desired rotation matrix from the orbit frame to the body frame $R^{current}_{orbit_body} = R^{desired}_{orbit_body}$ [1]:

$$\begin{aligned} R^{current}_{orbit_body} &= R_x(\varphi)R_y(\theta)R_z(\psi) \\ &= \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\cos\psi & -\sin\theta \\ -\cos\varphi\sin\psi + \sin\varphi\sin\theta\cos\psi & \cos\varphi\cos\psi + \sin\varphi\sin\theta\sin\psi & \sin\varphi\cos\theta \\ \sin\varphi\sin\psi + \cos\varphi\sin\theta\cos\psi & -\sin\varphi\cos\psi + \cos\varphi\sin\theta\sin\psi & \cos\varphi\cos\theta \end{bmatrix} \end{aligned} \tag{3}$$

$$R^{desired}_{orbit_body} = \begin{bmatrix} x_b \cdot x_o & x_b \cdot y_o & x_b \cdot z_o \\ y_b \cdot x_o & y_b \cdot y_o & y_b \cdot z_o \\ z_b \cdot x_o & z_b \cdot y_o & z_b \cdot z_o \end{bmatrix} \triangleq \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{4}$$

$$\begin{cases} \varphi_d = \operatorname{atan2}\dfrac{y_b \cdot z_o}{z_b \cdot z_o} \\ \theta_d = \arcsin(-x_b \cdot z_o) \\ \psi_d = 0 \end{cases} \tag{5}$$

[1] We refer the reader to the analysis in [37] and [14].

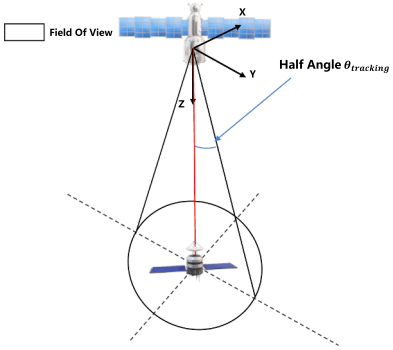


Fig. 3. Field of View of the onboard payload.

Then the corresponding desired quaternion $Q_d = [q_1^d, q_2^d, q_3^d, q_4^d]^T$ can be derived via:

$$\begin{cases} q_4^d = \frac{1}{2}(1 + a_{11} + a_{22} + a_{33})^{\frac{1}{2}} \\ q_1^d = \frac{1}{4q_4^d}(a_{23} - a_{32}) \\ q_2^d = \frac{1}{4q_4^d}(a_{31} - a_{13}) \\ q_3^d = \frac{1}{4q_4^d}(a_{12} - a_{21}) \end{cases} \tag{6}$$

Finally, the error quaternion Q_e is the quaternion relating the current body frame Q of the satellite to the desired frames Q_d, which can be obtained as follows:

$$Q_e = Q_d^{-1} \otimes Q \tag{7}$$

where $Q_d^{-1} = [-q_1^d, -q_2^d, -q_3^d, q_4^d]^T$ means the inverse or conjugate of the desired quaternion Q_d, and $\otimes$ means the quaternion multiplication operator of two quaternion.

The equation $Q_e = [0, 0, 0, 1]^T$ equals $Q = Q_d$, which means the agile satellite attitude reaches the desired attitude (payload sightline aligns with $\overrightarrow{ST}$) via attitude maneuvers.

3.1.3. Target tracking stage

In the final phase, the satellite executes fine attitude maneuvers to maintain continuous coverage within the limited visible time window, incorporating fault compensation strategies based on the earlier identification results. Following the large-angle attitude adjustment, the agile satellite successfully acquires the space target within its payload's field of view, as illustrated in Fig. 3.

Given the orbital dynamics, both the agile satellite and the space target undergo rapid positional changes, resulting in a time-varying desired quaternion Q_d. During the tracking stage, the satellite is required to execute fine attitude adjustments, which are crucial for maintaining continuous FOV coverage. Moreover, maintaining stable tracking is further achieved by keeping the angular velocity at a minimal level.

$$\begin{cases} \arccos \dfrac{\overrightarrow{ST}_{orbit} \cdot \hat{z}}{\left|\overrightarrow{ST}_{orbit}\right| \cdot |\hat{z}|} < \theta_{tracking} \\ \omega_i \le \omega_{tracking}, \forall \omega_i \in [\omega_x, \omega_y, \omega_z] \end{cases} \tag{8}$$

where $\theta_{tracking}$ denotes the half angle of onboard payload, $\omega_{tracking}$ denotes the acceptable max angular velocity during tracking, and $\hat{z}$ represents the transformed z-axis of satellite after the body-to-orbit frame rotation.

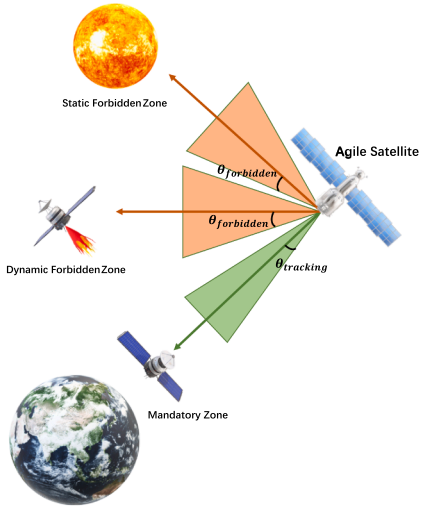


Fig. 4. Attitude Constraints in Ref [18].

3.2. Operation constraints

3.2.1. Attitude constraint

As extensively analyzed in [16–19] and visually represented in Fig. 4, the satellite's attitude control is subject to two principal constraint categories: attitude-forbidden zones and attitude-mandatory zones. The attitude-forbidden zones encompass both static and dynamic forbidden zones. The static forbidden zone is primarily associated with bright celestial bodies, while the dynamic forbidden zone typically corresponds to the thrust engines of neighboring spacecraft. These zones pose significant risks to the onboard payload, particularly imaging observation instruments, as exposure to intense radiation and brightness levels may cause irreversible damage or even complete payload system failure. Regarding the attitude-mandatory zone, it represents a specific attitude set that must be maintained to ensure the space target remains within the payload's field of view (FOV). This zone is crucial for successful target acquisition and continuous tracking operations.

Attitude-Forbidden Zones: Let $S_{forbidden} = \{S_{sf}, S_{df}\}$ represent the comprehensive set encompassing both static and dynamic forbidden sources, where S_{sf} denotes the collection of static forbidden sources and S_{df} represents the set of dynamic forbidden sources. To ensure compliance with the attitude-forbidden constraint, the angular separation between the payload axis $\hat{z}$ and the vectors from the agile satellite to each forbidden source ζ_f must exceed the specified forbidden sensitive angle $\theta_{forbidden}$. This constraint can be mathematically expressed as:

$$\arccos\left(\frac{\overrightarrow{S\zeta_{f,i}} \cdot \hat{z}}{\|\overrightarrow{S\zeta_{f,i}}\| \cdot \|\hat{z}\|}\right) > \theta_{forbidden}, \quad \forall \zeta_{f,i} \in S_{forbidden} \tag{9}$$

where $\zeta_{f,i}$ denotes the i-th attitude-forbidden source in the set $S_{forbidden}$, $\theta_{forbidden}$ represents the minimum acceptable angular separation threshold.

Attitude-Mandatory Zones: The operational regulation of the attitude-mandatory zone follows the same criterion as the tracking condition defined in Eq. 8. This ensures that the target remains within the field of view coverage of the onboard payload.

$$\arccos \frac{\overrightarrow{ST}_{orbit} \cdot \hat{z}}{\left|\overrightarrow{ST}_{orbit}\right| \cdot |\hat{z}|} < \theta_{tracking} \tag{10}$$

3.2.2. Angular velocity constraint

Considering the limited measurement range of gyroscopes and the operational requirements of space missions [19], the angular velocity ω

of the agile satellite is subject to the following constraint [2]:

$$\omega_i \leq \omega_{max}, \forall \omega_i \in [\omega_x, \omega_y, \omega_z] \tag{11}$$

where $\omega_{\max}$ represents the maximum allowable angular velocity magnitude to ensure stable and precise attitude control.

3.2.3. Torque constraint

Actuator saturation is a well-known challenge in traditional satellite attitude control design. This built-in constraint handling enhances the practicality and safety of the controller in real-world applications. Specifically, this design allows the agile satellite agent to make control torque decisions within a predefined and physically feasible range, inherently satisfying the actuator saturation constraint [21,28,29]:

$$|u_{ci}| \leq u_{max} \tag{12}$$

where u_{ci} denotes the command torque of any actuator along a given direction, and $u_{\max}$ is a positive constant.

3.3. Actuator faults

The precise attitude control of agile satellites critically depends on reliable actuator operation, with reaction wheels being particularly vulnerable to performance degradation. These high-precision mechanisms are prone to developing various anomalies across their electrical, mechanical, and power delivery subsystems due to prolonged operational stresses [21]. Empirical observations reveal that the predominant failure mechanisms stem from cumulative wear effects, including but not limited to material fatigue, lubrication breakdown, and increasing mechanical resistance. These faults manifest as either multiplicative effectiveness loss (e_l) or additive bias (u_a), leading to slower response times, reduced efficiency, or even complete failure of the reaction wheel. As done in previous works [32,38,39], actuator faults are considered in this study, reflecting the complexity and realism of actual tracking scenarios.

The attitude dynamics and kinematics models of the satellite with actuator faults are as follows:

$$\begin{cases} J\dot{\omega} = -\omega^{\times}(J \cdot \omega) + u + d \\ \dot{q} = \frac{1}{2}(q^{\times} + q_4 I_3)\omega \\ \dot{q}_4 = -\frac{1}{2} q^T \omega \end{cases} \tag{13}$$

where J is the inertia matrix, ω is the angular velocity of the agile satellite, $u \in \mathbb{R}^n$ denotes the actual torque, $d \in \mathbb{R}^3$ denotes the disturbance torque, $Q = [q_1, q_2, q_3, q_4]^T = [q, q_4]^T$ denotes the quaternion of body frame, and the notation $\times$ is used to represent the skew-symmetric cross-product matrix.

When a fault occurs, the actual torque u produced by the actuators deviates from the command torque u_c due to effectiveness loss e_l and additive bias u_a.

$$u = (I - E)u_c + u_a \tag{14}$$

where I is the identity matrix, $E = diag(e_{l1}, e_{l2}, \cdots, e_{\ln})$ represents the effectiveness loss of each actuator with $e_{li} \in [0, 1]$, $u_c = [u_{c1}, u_{c2}, \cdots, u_{cn}]^T$ is the command torque, and $u_a = [u_{a1}, u_{a2}, \cdots, u_{an}]^T$ is the additive bias fault. Substituting the fault model (Eq. 14) into the dynamics model (Eq. 13), the attitude dynamics considering the actuator faults can be written as:

$$\begin{aligned} J\dot{\omega} &= -\omega^{\times}(J \cdot \omega) + (I - E)u_c + u_a + d \\ &= -\omega^{\times}(J \cdot \omega) + u_c + f + d \end{aligned} \tag{15}$$

where $f = -Eu_c + u_a$ denotes the total effect of actuator faults on the spacecraft dynamics system.

[2] The constraint criterion differs from Eq. 8 ($\omega_{tracking} \ll \omega_{\max}$), because the latter is designed to ensure stable tracking.

4. Proposed method

This section presents the proposed fault-tolerant and safety-aware reinforcement learning framework for agile satellite-based non-cooperative space target tracking. The method consists of two key components: (i) a neural network-based fault identification module that estimates actuator effectiveness loss and additive bias, and (ii) a safe reinforcement learning controller based on a worst-case Soft Actor-Critic algorithm, which generates fault-tolerant attitude commands while ensuring safety limitations such as forbidden zone avoidance, angular velocity limits and torque saturation constraints. The overall framework enables intelligent non-cooperative target tracking in uncertain and fault-prone space environments.

4.1. Neural network for fault identification

In this paper, fault identification based on neural networks is framed as a time series regression problem. Specifically, in order to simplify the neural network's learning task by precomputing complex matrix operations, we re-analyze the spacecraft dynamics equation (Eq. 13) by moving the complex computation terms to the left-hand side,

$$J\dot{\omega} + \omega^{\times}(J \cdot \omega) = (I - E)u_c + u_a + d \tag{16}$$

and denote the left-hand side of this equation as the dynamics residuals Φ:

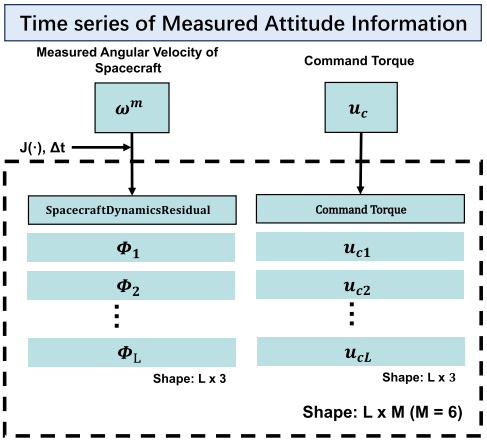
$$\begin{aligned} \Phi &= J\dot{\omega} + \omega^{\times}(J \cdot \omega) \\ &\approx J(\omega^m_{new} - \omega^m_{old})/dt + \omega^m_{old^{\times}}(J \cdot \omega^m_{old}) \end{aligned} \tag{17}$$

where ω^m_{new} and ω^m_{old} represent the measured angular velocity of the satellite at two consecutive time steps. As illustrated in Fig. 5, the neural network processes the time series in both the dynamics residuals and command torques. The input tensor has dimensions $L \times M$, where L represents the look-back window size, and $M = 6$ features consist of 3-axis dynamics residuals Φ and corresponding command torques u_c. The network outputs the estimated actuator faults with dimensions $1 \times N$, where $N = 6$ represents the complete fault information vector: 3-axis estimated effectiveness loss $\hat{e}_l$ and additive bias faults $\hat{u}_a$. As illustrated in Fig. 6, we adopt the similar architecture in [32] and enhance the attention layer with a convolutional block attention module (CBAM) to improve estimation accuracy. Specifically, in the feature extraction phase, a bidirectional LSTM structure (implemented by $nn.LSTM$) is adopted, with an input dimension of 6 ($input_size = 6$) and a hidden layer dimension set to 128 ($hidden_size = 128$). Due to the enabled bidirectional mode ($bidirectional = True$), the output feature dimension of the LSTM doubles to 256, forming a time-series feature matrix with the shape $[B, L, 256]$, where B is the batch size and L is the look-back window size. The attention enhancement layer is the core improvement point of this model, implementing a lightweight CBAM attention module, and its workflow is divided into two sub-modules: the channel attention module and the spatial attention module. This CBAM attention mechanism effectively identifies critical time steps as well as important spatial and channel-wise features within the temporal sequences via channel and spatial attention layers. After dimension conversion and mean operation, the output of the attention module is concatenated with the output of the LSTM to form a feature vector of $[B, 512]$. This design not only retains the global information of the time series but also incorporates local key features. In the output phase, the fully connected layer maps the 512-dimensional features to a 6-dimensional output. These output features pass through an activation layer, and finally, they are concatenated into an output tensor of $[B, 1, 6]$.

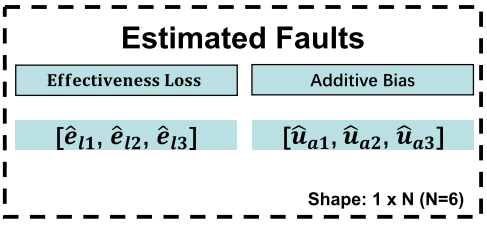
4.2. Safe reinforcement learning for tracking

4.2.1. Constrained markov decision process

As done in the most safe reinforcement learning applications [40,41], the interaction between the agile satellite agent and tracking environment is regarded as a Constraint Markov Decision Process [6]. A tuple

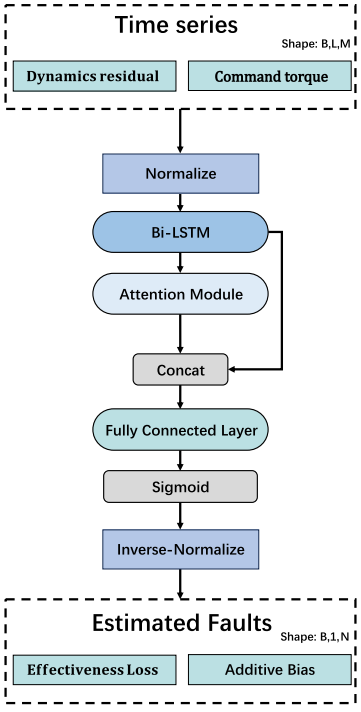


(a) Input of Neural Network

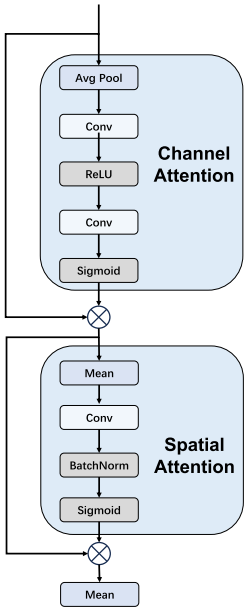


(b) Output of Neural Network

Fig. 5. Input and output of Neural Network.



(a) Network Architecture



(b) CBAM Attention Module

Fig. 6. Neural Network Architecture.

$M = (S, A, p, r, c, d_{cost}, \gamma)$ denotes a CMDP, where S is the state space, A is the action space, p is the state transition probability: $S \times A \times S' \Longrightarrow R$ represents the transition from state S to new state S' after executing action A, $r \in [r_{\min}, r_{\max}]$ is the reward after taking action, $c \in [c_{\min}, c_{\max}]$ is the cost after taking action, d_{cost} is given safety threshold, defining the maximum allowable cumulative cost, and $\gamma \in [0, 1)$ is the discount factor balancing the trade-off between immediate and future rewards.

The process unfolds as follows: At each discrete time step, the agent observes the current state $s \in S$. Based on S, the agent selects an action $a \in A$. The environment transitions to a new state $S' \in S$ according to the probability distribution p. The agent receives a reward $r(S, A, S')$ and a cost $c(S, A, S')$. This process repeats until a terminal state is reached. In a Constrained Markov Decision Process, the policy optimization objective and safety constraints are defined as follows:

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t, a_t, s_{t+1})\right] \\ \text{s.t.} \quad & \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t c(s_t, a_t, s_{t+1})\right] \le d_{cost} \end{aligned} \tag{18}$$

4.2.2. Worst-case SAC

To deal with the randomness in long-term costs, the worst-case soft actor-critic (WCSAC) algorithm [7] is applied by depicting the long-term costs with its current cost returns and variance via replacing the original single-output safety critic with a distributional-output critic. Specifically, as done in [7,42], the distributional critic is approximated with a Gaussian distribution.

$$C_\pi \sim N(Q_\pi^c(s, a), V_\pi^c(s, a)) \tag{19}$$

where Q_π^c and V_π^c represents the mean and variance of this distribution:

$$Q_\pi^c(s, a) = c + \gamma \sum_{s'\in S} p(s'|s, a) \sum_{a'\in A} \pi(a'|s') Q_\pi^c(s', a') \tag{20}$$

$$\begin{aligned} V_\pi^c(s, a) = {} & c^2 - Q_\pi^c(s, a)^2 \\ & + 2\gamma c \sum_{s'\in S} p(s'|s, a) \sum_{a'\in A} \pi(a'|s') Q_\pi^c(s', a') \\ & + \gamma^2 \sum_{s'\in S} p(s'|s, a) \sum_{a'\in A} \pi(a'|s') V_\pi^c(s', a') \\ & + \gamma^2 \sum_{s'\in S} p(s'|s, a) \sum_{a'\in A} \pi(a'|s') Q_\pi^c(s', a') \end{aligned} \tag{21}$$

Then the parameters of the safety critic network are updated via gradient descent using the following equation, which incorporates the

Temporal Difference (TD) error:

$$\begin{cases} L_{\text{safety critic}}(\mu,\eta) = L_Q(\mu) + L_V(\eta) \\ L_Q(\mu) = \underset{(s_t,a_t)\sim p_\pi}{E} ||\overline{Q}^c_{\pi,\mu}(s',a') - Q^c_{\pi,\mu}(s',a')||^2 \\ L_V(\eta) = \underset{(s_t,a_t)\sim p_\pi}{E} trace(\overline{V}^c_{\pi,\eta}(s',a') + V^c_{\pi,\eta}(s',a') \\ \qquad - 2(V^c_{\pi,\eta}(s',a')^{\frac{1}{2}}\overline{V}^c_{\pi,\eta}(s',a')V^c_{\pi,\eta}(s',a')^{\frac{1}{2}})^{\frac{1}{2}}) \end{cases} \tag{22}$$

where $L_Q(\mu)$ and $L_V(\eta)$ represent the TD errors loss function for the safety critic's mean $Q^c_{pi,\mu}$ and variance $V^c_{pi,\eta}$, respectively. Here, μ and η denote the trainable parameters of the respective neural networks, while the overline notation means the target value used in the TD error computation.

As for the actor network, the action is decided by taking the new safety measure $\Gamma_\pi(s,a,\alpha)$ into account to ensure safe exploration.

$$\Gamma_\pi(s,a,\alpha) = Q^c_\pi(s,a) + \frac{1}{\alpha}\phi(\Phi(\alpha))\sqrt{V^c_\pi(s,a)} \tag{23}$$

where $\Gamma_\pi(s,a,\alpha)$ is the new safety measure under risk level α under the distribution safety critic. The actor network should decide an action which satisfies the safety requirement $\Gamma_\pi(s,a,\alpha) \le d$.

The trainable parameter of the actor network is updated by minimizing the following equation:

$$\begin{cases} L_{\text{actor}}(\theta) = \underset{s\sim D, a\sim\pi_\theta}{E}[\beta\log(\pi_\theta(a_t|s_t)) - X^{\pi_\theta}_{\alpha,k}(s_t,a_t)] \\ X^{\pi_\theta}_{\alpha,k}(s_t,a_t) = Q^r_\pi(s,a) - k\Gamma_\pi(s,a,\alpha) \\ \qquad\qquad = Q^r_\pi(s,a) - k(Q^c_\pi(s,a) + \frac{1}{\alpha}\phi(\Phi(\alpha))\sqrt{V^c_\pi(s,a)}) \end{cases} \tag{24}$$

where β represents the adaptive temperature weight in soft actor critic [43,44] to balance exploration and exploitation, and k represents the adaptive safety cost weight similar in SACLag [45] and SACPID [46] to balance safety and cost. Both of the above weight parameters can be updated via gradient descent to achieve a better balance.

$$\nabla_\beta J(\beta) = E(-\beta(log\pi(a|s) + H)) \tag{25}$$

$$\nabla_k J(k) = E(k(c_{target} - \Gamma_\pi(s,a,\alpha))) \tag{26}$$

where c_{target} denotes the cost target of WCSAC.

In a word, the WCSAC algorithm essentially constitutes a specialized variant of SACLag, wherein the conventional single-output safety critic network is superseded by a distributional alternative to improve the estimation accuracy in long-term cost.

4.2.3. Interaction between RL algorithm and environment

In the context of agile satellite-based space target tracking, the agile satellite is required to continuously track a moving non-cooperative space target while ensuring that its attitude maneuver does not violate forbidden orientation zones. In addition, angular velocity is subject to operational limitations, and actuator faults may occur during operation. The interaction loop between the RL agent and the environment is illustrated in Fig. 7. At each decision step, the RL agent (based on a worst-case Soft Actor-Critic algorithm) receives the current state of the system, which includes target and forbidden zone orientation quaternion errors, measured angular velocity, and estimated actuator fault information. Based on this input, the agent outputs a command torque as the action. This action is executed on the satellite dynamics, which considers both system disturbances and actuator faults, including effectiveness loss and additive bias. The updated satellite attitude and angular velocity are used to compute the new state, reward, and cost, which are then fed back to the RL agent for policy updates.

State Space: The state vector $s \in S$ encompasses a comprehensive description of the satellite's current attitude information and actuator faults, enabling effective policy learning under operation constraints (attitude and angular velocity constraints) and actuator faults (effectiveness loss and additive bias). The state is defined as:

$$s = \left[Q_{T,e},\ Q_{Z_1,e},\cdots,Q_{Z_N,e},\ \omega^m,\ \hat{e}_l,\ \hat{u}_a\right] \tag{27}$$

where:

- $Q_{T,e} = [q_1^{T,e}, q_2^{T,e}, q_3^{T,e}, q_4^{T,e}]^T$: Error quaternion between the current satellite attitude and the target pointing direction.
- $Q_{Z_i,e} = [q_1^{Z_i,e}, q_2^{Z_i,e}, q_3^{Z_i,e}, q_4^{Z_i,e}]^T$: Error quaternions between the satellite attitude and the i-th forbidden zone orientation, for $i = 1,\ldots,N$, where N denotes the number of forbidden zones.
- ω^m: Measured angular velocity vector of the satellite.
- $\hat{e}_l$: Estimated actuator effectiveness loss vector from the fault identification stage.
- $\hat{u}_a$: Estimated actuator additive bias from the fault identification stage.

Our state representation adopts a unified error quaternion formulation that inherently encodes the relative geometric relationships between the spacecraft's attitude, target positions, and forbidden zones, enabling the agent to simultaneously optimize tracking performance while maintaining safety constraints and compensating for actuator degradation in a generalized manner. This error quaternion-based state representation fundamentally decouples the control policy from specific orbital configurations and improves generalization capabilities, allowing consistent performance across varying mission scenarios - when target locations shift due to orbital mechanics, when new forbidden zones are introduced operationally, or when actuator performance degrades mid-mission.

Action Space: The action space $a \in \mathcal{A}$ is defined as the commanded control torque applied to the satellite body, subject to actuator constraints. That is:

$$a = \tau_c \in \mathbb{R}^3 \tag{28}$$

where τ_c denotes the 3-axis command torque output by the agent. In the presence of actuator faults, the actual torque executed by the satellite differs due to loss of effectiveness and additive bias, modeled in the environment dynamics. In addition, by defining this torque bound in the RL framework, the control strategy ensures that the resulting control signals always respect the physical torque constraint (as shown in Eq. 12).

Reward Function: The control objective is twofold: to minimize the attitude error relative to the target–driving the target error quaternion $Q_{T,e}$ toward $[0,0,0,1]^T$–and to maximize the deviation from forbidden pointing directions by ensuring the forbidden zone error quaternion $Q_{Z_i,e}$ remains sufficiently far from $[0,0,0,1]^T$. The reward function is designed to guide the agent toward achieving accurate and stable target tracking. It includes multiple components:

$$r_{sum} = k_1 \cdot r_{maneuver} + k_2 \cdot r_{track} + k_3 \cdot r_{stability} + k_4 \cdot r_{safety} \tag{29}$$

where k_1, k_2, k_3, and k_4 represent the weights assigned to each individual reward, respectively, and the subscripts denote the specific reward components:

$$\begin{cases} r_{\text{maneuver}} = q_4^e \\ r_{\text{track}} = \begin{cases} 9, & \text{if target in the FOV} \\ 0, & \text{otherwise} \end{cases} \\ r_{\text{stability}} = \begin{cases} -(|\omega_x| + |\omega_y| + |\omega_z|), & \text{if target in the FOV} \\ 0, & \text{otherwise} \end{cases} \\ r_{\text{safety}} = \begin{cases} -15, & \text{if} \omega_i > \omega_{tracking}, \forall\omega_i \in [\omega_x,\omega_y,\omega_z] \\ 0, & \text{otherwise} \end{cases} \end{cases} \tag{30}$$

- **Maneuver Reward (r_{maneuver})**: This term uses the scalar part q_4^e of the attitude error quaternion $Q_{T,e}$ to encourage alignment between the satellite's sightline and the target direction. A larger q_4^e implies better alignment, hence a higher reward.
- **Tracking Reward (r_{track})**: A discrete reward of 9 is provided when the target is successfully located within the onboard payload's field-of-view, thereby indicating the satellite has achieved a valid tracking status. Otherwise, this term returns zero.

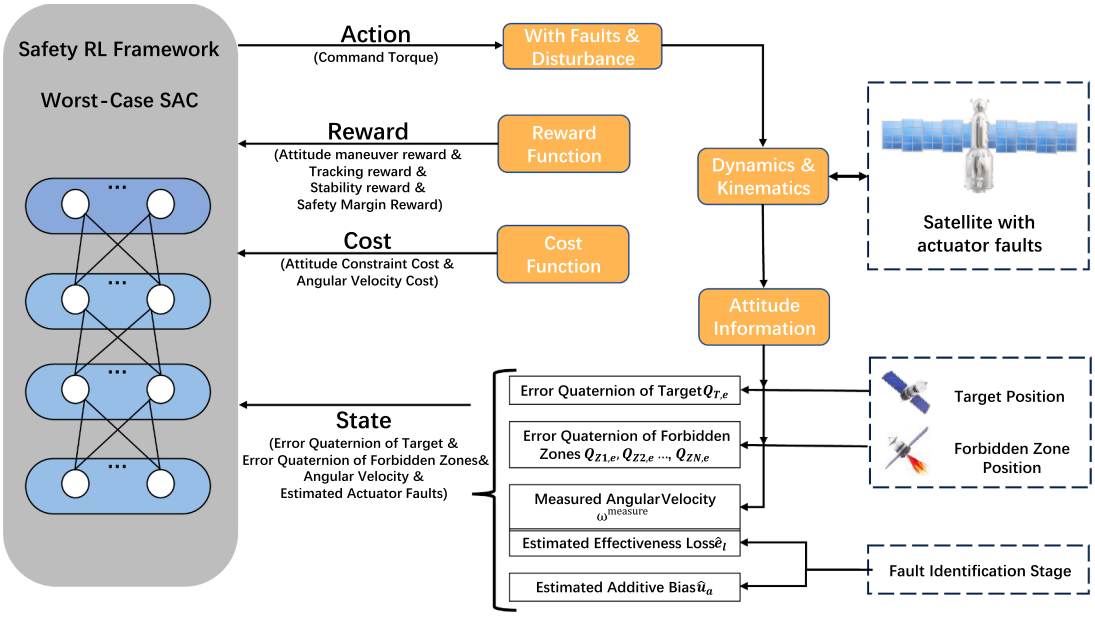


Fig. 7. Interaction between RL algorithm and Environment.

- **Stability Reward ($r_{\text{stability}}$)**: This penalty term discourages high angular velocity magnitudes during tracking. Excessive rotational rates may destabilize tracking; hence, the total angular velocity is penalized when the target is being tracked.
- **Safety Penalty (r_{safety})**: This hard penalty is triggered when any angular velocity component exceeds a predefined safety threshold, which could otherwise cause mechanical damage or loss of payload integrity.

Cost Function: The cost function is designed to quantify safety violations during the satellite's tracking maneuvers, particularly in terms of forbidden zone avoidance and angular velocity constraints. The total cost c at each step is expressed as:

$$c = c_{\text{attitude}} + c_{\omega} \tag{31}$$

where the two components are defined as:

$$\begin{cases} c_{\text{attitude}} = \sum_{i=1}^{N} c_{attitude,i} \\ c_{\omega} = \begin{cases} 1, & \text{if} \|\omega\| \le \omega_{max} \\ 0, & \text{otherwise} \end{cases} \end{cases} \tag{32}$$

The terms are defined as:

- **Attitude Constraint Cost (c_{attitude})**: This component penalizes violations of forbidden orientation zones. For each forbidden zone i, the satellite checks whether the zone lies within the payload's field-of-view, as illustrated in Eq. 9. A discrete cost of 1 is incurred if the i-th forbidden zone constraint was violated:

$$c_{attitude,i} = \begin{cases} 1, & \text{if} \arccos\left(\frac{\overrightarrow{Sc_{f,i}} \cdot \hat{z}}{\|\overrightarrow{Sc_{f,i}}\| \cdot \|\hat{z}\|}\right) \le \theta_{forbidden} \\ 0, & \text{otherwise} \end{cases} \tag{33}$$

- **Angular Velocity Cost (c_{ω})**: A discrete cost of 1 is incurred if any component of the angular velocity vector exceeds the predefined angular velocity threshold, as illustrated in Eq. 11. A discrete cost of 1 is returned if $\omega_i > \omega_{\max}, \forall \omega_i \in [\omega_x, \omega_y, \omega_z]$.

4.3. Pseudo-code

The learning and deployment process of the proposed safe reinforcement learning framework for non-cooperative space target tracking is summarized in Algorithm 1. The training phase achieves comprehensive trajectories by randomly generating initial pose conditions and fault scenarios. Interaction trajectories are sampled using a replay buffer strategy with a length of 10^6. During training, the agile satellite agent learns to track the moving target while respecting safety constraints under various actuator fault conditions. The agent receives full fault information during training to encourage fault-tolerant behavior. In contrast, during testing, actuator faults are estimated through a pre-trained fault identification network, and the agent makes decisions based on estimated faults.

At each timestep, the agent observes the current state, which includes tracking errors, forbidden zone orientation errors, measured angular velocity, and fault information. The agent outputs a control torque command based on the policy learned using a worst-case Soft Actor-Critic algorithm. The environment simulates actuator-impaired satellite dynamics, and returns both a reward signal and a safety cost. These experience tuples are stored in a replay buffer to improve sample efficiency. The actor and critic networks are updated periodically based on a mini-batch sampled from the buffer. Both reward and cost critics are updated using temporal-difference learning, and the temperature and cost weights are also updated via gradient descent, thus ensuring safety during policy learning.

5. Experiment

5.1. Implementation configuration & scenario simulation

The computational experiments were performed on a high-performance computing platform featuring an AMD EPYC 7453 CPU and NVIDIA RTX 4090 GPU, running the Ubuntu 22.04 operating system. The software environment comprised PyTorch 2.2.2 [47] with CUDA 12.3 acceleration, complemented by Omnisafe reinforcement learning package [48] and Python 3.11.8 interpreter. Considering the lightweight requirements for spaceborne missions, both the neural network for WC-SAC used in safe attitude control and the neural network for fault diagnosis and identification were trained within a few hours–a reasonable timeframe for intelligent spacecraft system.

The target tracking simulation environment was developed using Omnisafe's Constrained Markov Decision Process interface, providing a standardized framework for safe reinforcement learning research. In

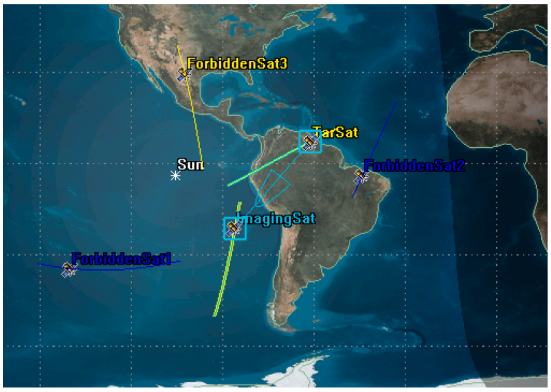
Algorithm 1: Safe RL Framework for Target Tracking.

if *Mode* $==$ *'train'* **then**
  Randomly initialize the quaternion Q_{init} and angular velocity ω_{init} of the satellite agent;
  Randomly initialize the values of actuator faults.;
  # No fault estimation in training; true faults are directly available ;
  $\hat{e}_l \leftarrow e_l$;
  $\hat{u}_a \leftarrow u_a$;
else if *Mode* $==$ *'test'* **then**
  Initialize the pre-defined quaternion Q_{init} and angular velocity ω_{init} of the satellite agent based on METADATA;
  Initialize the pre-defined values of actuator faults based on METADATA.;
  # Fault Identification via pre-trained neural network ;
  Apply nominal torque input u_c;
  $(\hat{e}_l, \hat{u}_a) \leftarrow \text{Network}(\Phi, u_c)$;
Receive initial state s_1;
for $t = 1$ *to Terminal timestep* **do**
  Choose action a_t based on RL algorithm;
  Receive reward r and cost c based on reward and cost function;
  update the attitude of satellite agent based on Eq. (13).;
  Collect attitude and fault information to compose a new state s_{t+1};
  Store $(s_t, a_t, r_t, c_t, s_{t+1})$ in Replay Buffer;
  $s_t \longleftarrow s_{t+1}$
if *update interval reached* **then**
  Randomly select a batch of transitions $\{s_i, a_i, r_i, c_i, s_{i+1}\}_{i=1,2,\ldots,N}$ from Replay Buffer.
  Update reward critic and cost critic using TD targets;
  Update actor network via policy gradient;
  Update the temperature weight β and safety cost weight k.

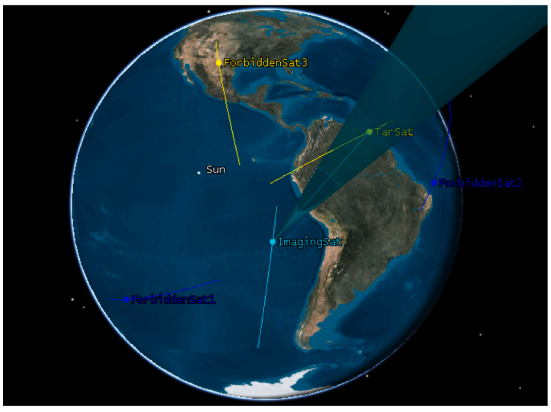
Table 1
Stage duration.

Stage	Duration
Fault Identification	1 Oct 2024 18:59:57.000 - 18:50:00.000 (3s)
Attitude Adjustment & Target Tracking	1 Oct 2024 18:50:00.000 - 18:59:00.000 (540s)

our approach, the constraints are primarily implemented as soft constraints through adjustable penalty terms in the cost function, which allows the agent policy to learn constraint satisfaction while maintaining reward optimization. As shown in Table 1, each experimental trial followed a structured timeline: an initial 3-second phase dedicated to fault identification and parameter estimation, followed by a 540-second operational period for attitude maneuver execution and target tracking. The time step for fault identification and attitude control is set to 0.1s and 1s, respectively. Fig. 8 illustrates the non-cooperative space target tracking scenario from 2d and 3d aspects, respectively. The ImagingSat (satellite agent) should continuously track the TargetSat (space target) while ensuring the operation constraints and compensating actuator faults. The spacecraft attitude dynamics are numerically integrated using a 4th-order Runge-Kutta (RK4) method. Specifically, the tracking scenario has four forbidden zones for testing attitude constraints, including one stable forbidden zone (Sun) and three dynamic forbidden zones (ForbiddenSat). During each timestep iteration (i.e., at every `step()` call of the reinforcement learning environment), the RL environment performs comprehensive operation constraint checking. All orbital elements and ephemeris data are computed using professional Systems



(a) 2D Visualization of Tracking Scenario



(b) 3D Visualization of Tracking Scenario

Fig. 8. Non-cooperative Space Target Tracking Scenario.

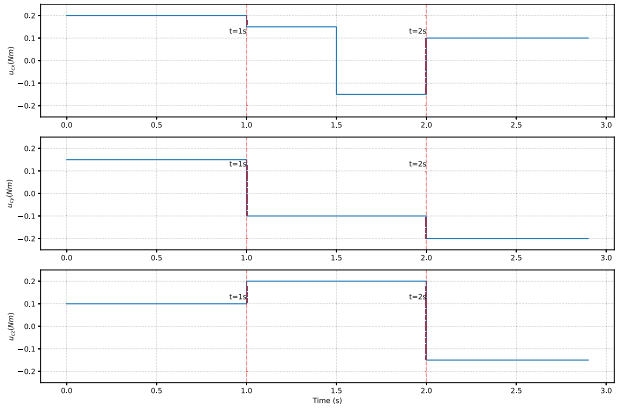


Fig. 9. Nominal torques during fault identification stage.

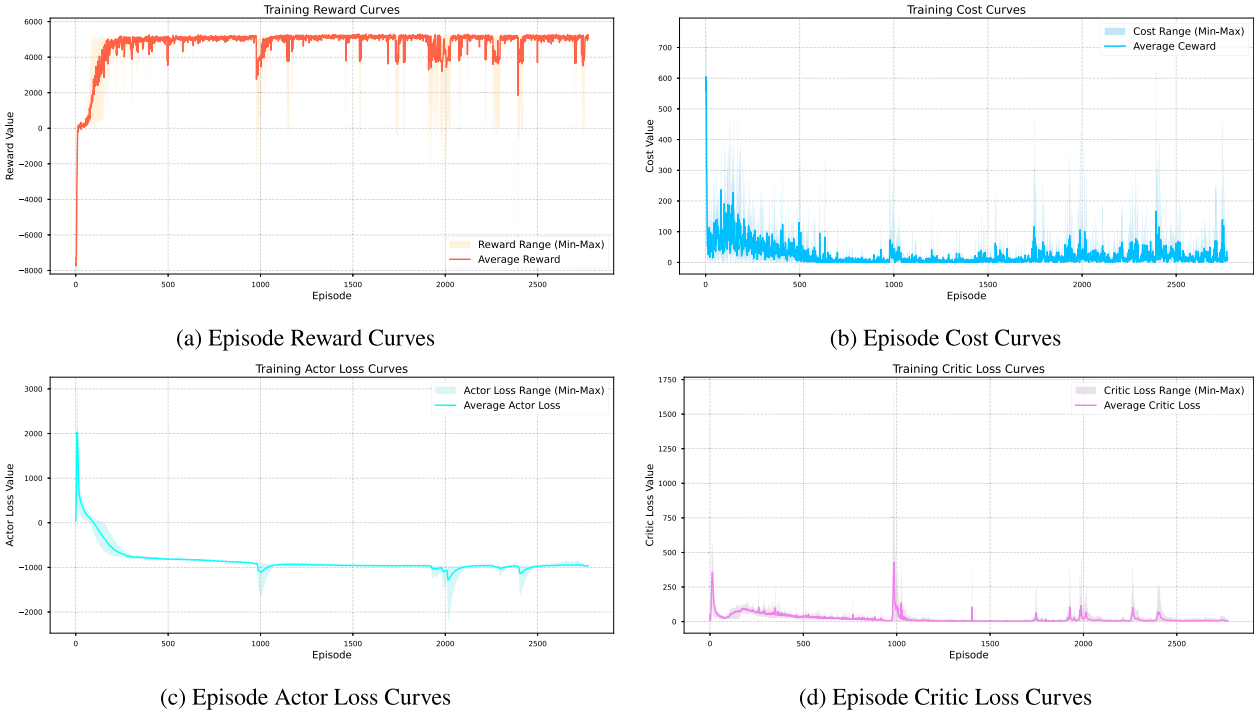
Tool Kit software based on their orbital elements. The detailed orbit elements of satellites in the tracking scenario are shown in Table 2.

Fig. 9 illustrates the detailed nominal torques during the fault identification stage. The layers of the neural network architecture used for identification are implemented by Torch [47]. The spacecraft's dynamics parameters, including the inertia matrix J and total disturbance torques d, as well as the measurement noise of angular velocity, are defined as follows:

$$J = \begin{bmatrix} 10 & 0.02 & 0.01 \\ 0.02 & 15 & -0.01 \\ 0.01 & -0.01 & 20 \end{bmatrix} kg \cdot m^2 \tag{34}$$

Table 2
Orbit elements of satellites.

Orbit Elements		Semimajor axis(km)	Eccentricity	Inclination(deg)	Argument of Perigee(deg)	RAAN(deg)	True Anomaly(deg)
Satellites	ImagingSat	6900	0	96	150	30	7
	TargetSat	6785	0.0169	150	45	40	0
	ForbiddenSat1	6511	5.0e-15	35	0	255	210
	ForbiddenSat2	7321	0.013	62	9.5e-12	252	312
	ForbiddenSat3	6421.1	4.9e-15	77	0	19	98



(a) Episode Reward Curves

(b) Episode Cost Curves

(c) Episode Actor Loss Curves

(d) Episode Critic Loss Curves

Fig. 10. Episode Curves of WCSAC during Training.

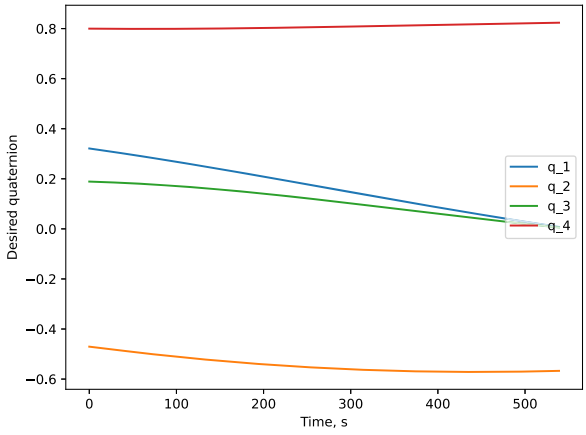


Fig. 11. Desired Quaternion Calculated by Eq. 6.

Table 3
Parameter Configuration.

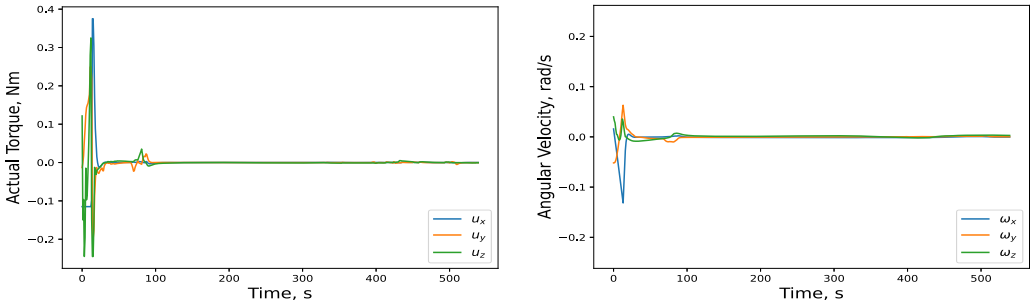
Parameter		Value
Satellite Agent	command torque range	$[-0.5, 0.5]$ Nm
	effectiveness loss	rand($[0, 1]$)
	additive bias	rand($[-0.15, 0.15]$) Nm
	max angular velocity for tracking	$\pi/12$ rand/s
	max angular velocity for constraint	$\pi/10$ rand/s
	payload half angle for tracking $\theta_{tracking}$	10 deg
	payload half angle for constraint $\theta_{forbidden}$	25 deg
WCSAC	safety threshold d_{cost}	5
	discount factor γ	0.99
	risk level α	0.8
	$k1, k2, k3, k4$	$1, 1, 10, 1$
	initial value of weight α	0.6931
	initial value of weight k	0.6931
	learning rate	0.001
	replay buffer size	1×10^6
	total timesteps for training	3×10^6
	optimizer	Adam

$$d = 10^{-5}\begin{bmatrix} -10 + 4sin(3\omega t) + 3cos(10\omega t) \\ 15 + 1.5sin(2\omega t) + \cos(5\omega t) \\ 10 + 3sin(10\omega t) + 8\sin(4\omega t) \end{bmatrix} Nm \tag{35}$$

$$\omega^m = \omega + \eta_{noise} \tag{36}$$

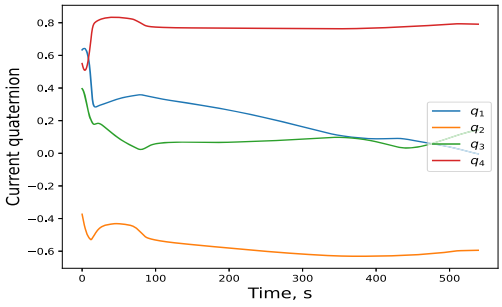
where the measurement noise component η_{noise} is a hybrid noise, similar to the model presented in Ref. [32,49]. It consists of two independent random signals, both with a mean of zero. The first component has a standard deviation of 0.1 % of the magnitude of ω, while the second component has a standard deviation of 0.01 % of the magnitude of the previous ω. Thus, the noise model can be expressed as:

$$\begin{cases} \eta_{noise} = \eta_{noise,1} + \eta_{noise,2}, \\ \eta_{noise,1} \sim \mathcal{N}(\mu = 0, \sigma^2 = (0.001|\omega|)^2), \\ \eta_{noise,2} \sim \mathcal{N}(\mu = 0, \sigma^2 = (0.0001|\omega_{\text{old}}|)^2) \end{cases} \tag{37}$$

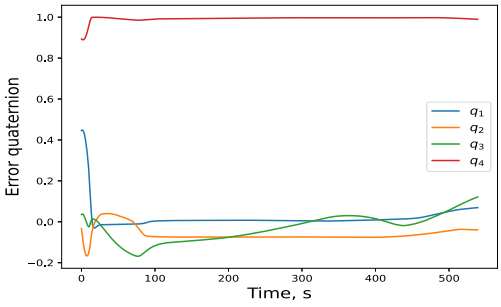


(a) Actual Torque (Case1)

(b) Angular Velocity (Case1)

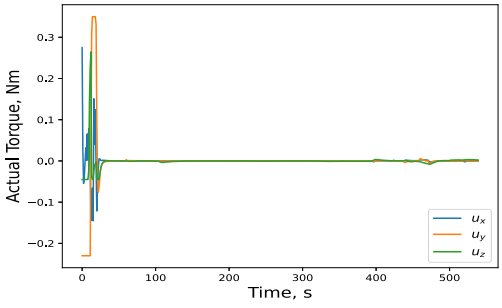


(c) Current Quaternion (Case1)

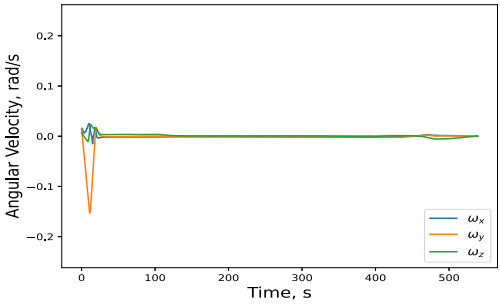


(d) Error Quaternion (Case1)

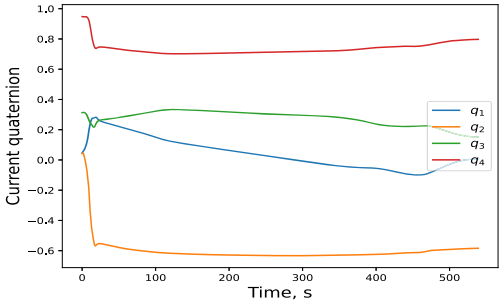
Fig. 12. Dynamics responses illustration (Case 1).



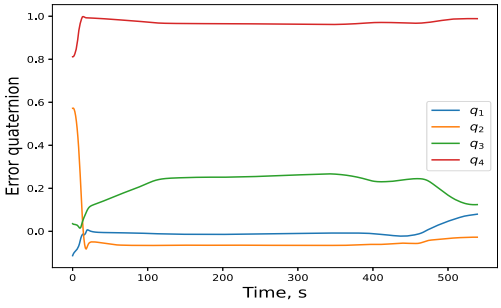
(a) Actual Torque (Case2)



(b) Angular Velocity (Case2)



(c) Current Quaternion (Case2)

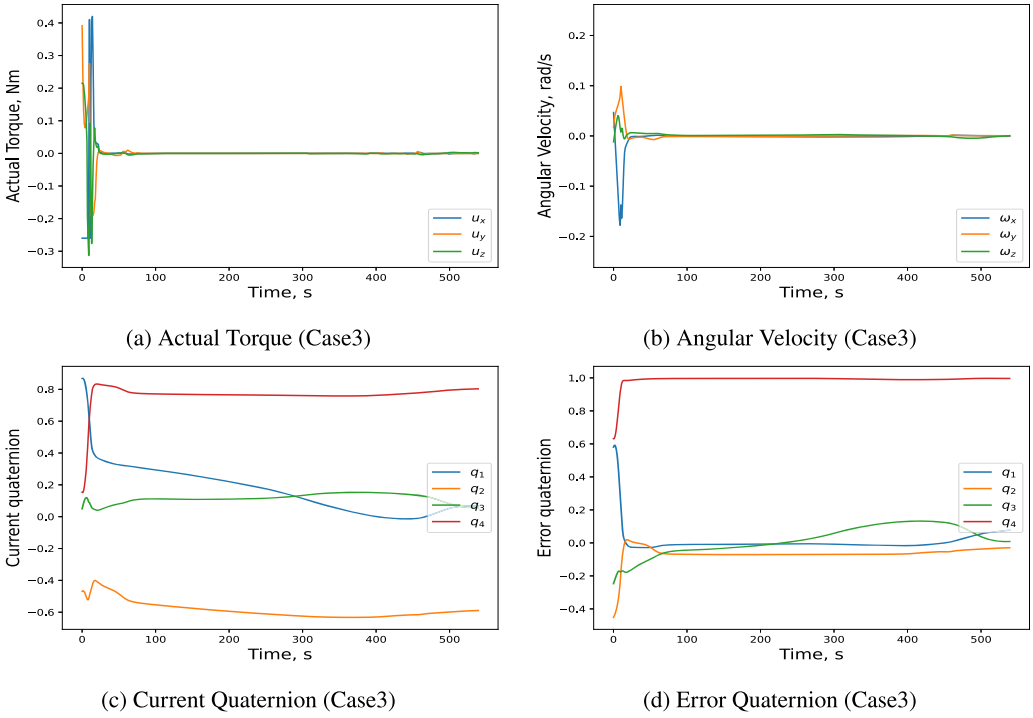


(d) Error Quaternion (Case2)

Fig. 13. Dynamics responses illustration (Case 2).

Table 4
Initial attitude information and actuator faults in five testing cases.

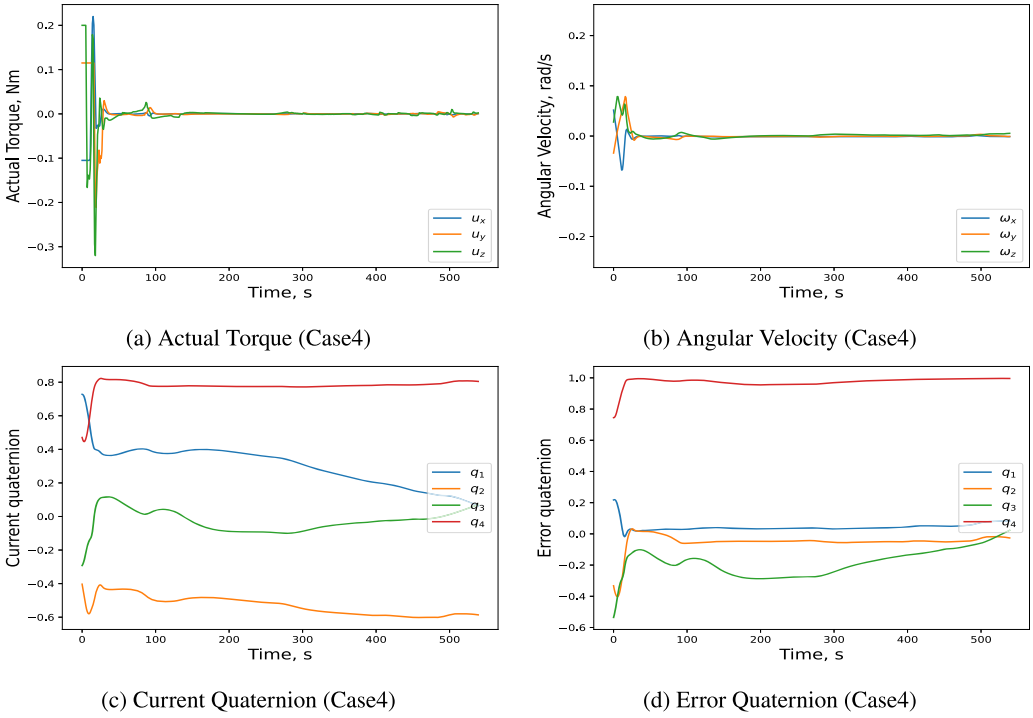
Testing Case	Initial Attitude Information		Initial Actuator Faults	
	Initial Quaternion	Initial Angular Velocity(rand/s)	Effectiveness Loss	Additive Bias(Nm)
Case 1	$[0.4611, 0.2515, 0.2774, 0.8044]^T$	$[-0.0290, -0.0268, 0.0231]^T$	$[0.51, 0.32, 0.43]^T$	$[0.13, -0.09, 0.04]^T$
Case 2	$[0.3003, -0.3849, 0.2489, 0.8364]^T$	$[-0.014, 0.0248, -0.0147]^T$	$[0.31, 0.42, 0.63]^T$	$[-0.07, 0.06, 0.14]^T$
Case 3	$[0.3320, -0.4676, -0.1528, 0.8047]^T$	$[0.0286, -0.0245, -0.0180]^T$	$[0.32, 0.38, 0.45]^T$	$[0.08, 0.12, -0.06]^T$
Case 4	$[0.3778, 0.2672, -0.3498, 0.8145]^T$	$[0.0222, -0.0134, 0.0216]^T$	$[0.61, 0.51, 0.48]^T$	$[0.09, -0.13, -0.06]^T$
Case 5	$[0.1584, 0.3066, 0.3303, 0.8784]^T$	$[-0.0132, -0.0158, 0.0112]^T$	$[0.33, 0.67, 0.63]^T$	$[-0.04, -0.05, 0.07]^T$



(a) Actual Torque (Case3) (b) Angular Velocity (Case3)

(c) Current Quaternion (Case3) (d) Error Quaternion (Case3)

Fig. 14. Dynamics responses illustration (Case 3).



(a) Actual Torque (Case4) (b) Angular Velocity (Case4)

(c) Current Quaternion (Case4) (d) Error Quaternion (Case4)

Fig. 15. Dynamics responses illustration (Case 4).

In addition, simulation parameters in this paper and algorithm parameters of WCSAC are illustrated in Table 3.

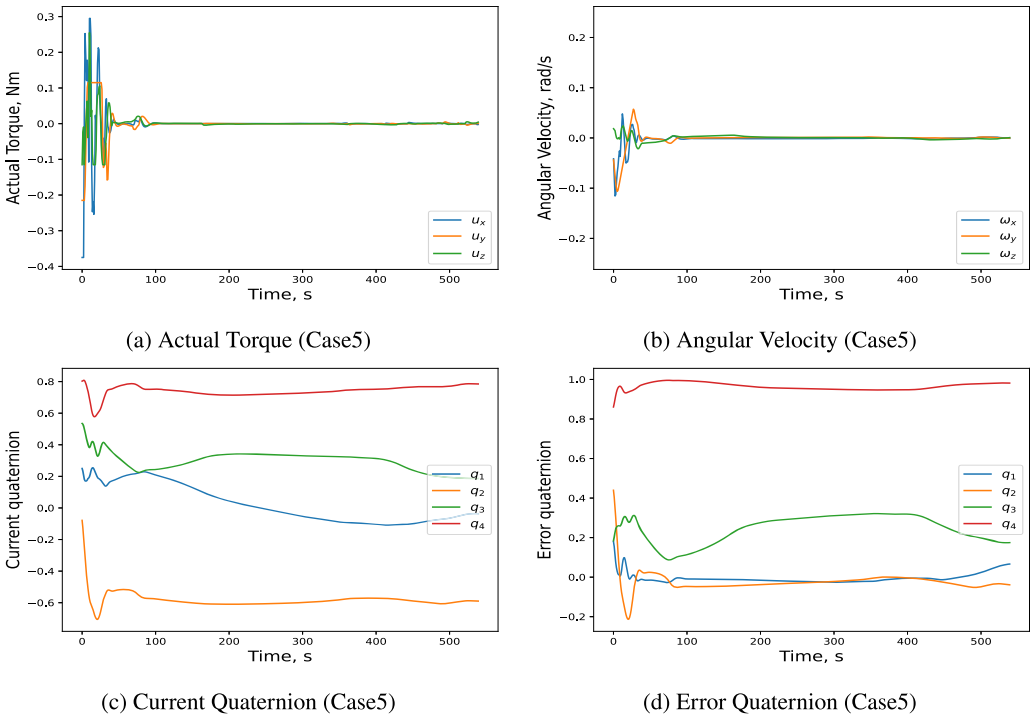
5.2. Comparison experiment

To validate the effectiveness and safety of the proposed fault-tolerant reinforcement learning framework for space target tracking, we conduct a series of comparison experiments against several baseline algorithms under various actuator fault scenarios and ablation experiments to validate the proposed components.

5.2.1. Comparison methods

Specifically, we compare our method with the following representative algorithms:

- **SAC** [43,44]: The vanilla Soft Actor-Critic algorithm without any safety mechanism or fault awareness. Serves as a baseline for performance.



(a) Actual Torque (Case5)

(b) Angular Velocity (Case5)

(c) Current Quaternion (Case5)

(d) Error Quaternion (Case5)

Fig. 16. Dynamics responses illustration (Case 5).

- **SACLag** [45]: SAC with a Lagrangian-based constraint handling mechanism that introduces a cost critic and a cost penalty multiplier to enforce safety constraints during learning.
- **SACPID** [46]: Improved version of SACLag, where a proportional-integral-derivative (PID) controller updates the lagrangian multiplier.
- **TD3PID** [46]: Improved version of TD3Lag, where a proportional-integral-derivative controller updates the lagrangian multiplier.
- **Reward Penalty (RP)**: A reward-shaping method that introduces penalties in the reward function when safety violations (e.g., forbidden zone violations or excessive angular velocity) occur.

$$r_{RP} = r - \lambda_{RP} \cdot c \tag{38}$$

where r_{RP} denotes the returned reward of each timestep after the reward penalty operation, r and c represent the original reward and cost mentioned above, respectively, and $\lambda_{RP} = 5$ is the cost weight.

Regarding test case selection, five primary cases were systematically chosen to cover the complete spectrum of anticipated operating conditions. These cases aim to represent: 1) arbitrary satellite attitudes, 2) arbitrary spacecraft angular velocities, and 3) arbitrary actuator faults (including effectiveness loss $e_l \in [0, 1]$ and additional bias $u_a \in [-0.15, 0.15] Nm$). As listed in Table 4, the selection process considered all critical operational parameters, including satellite attitude conditions and potential fault combinations. The training and evaluation hyperparameters are kept consistent across all methods for fair comparison.

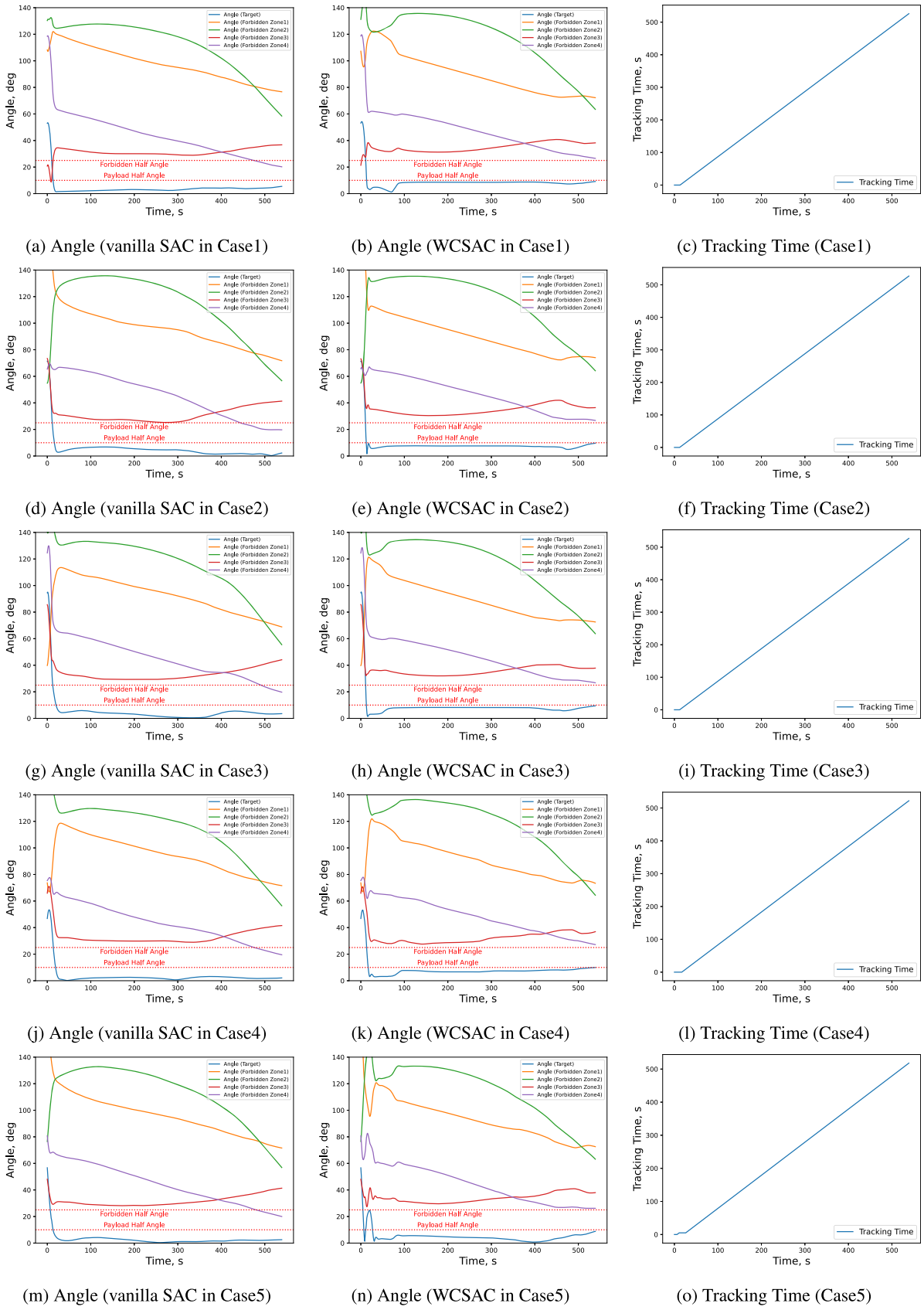
5.2.2. Performance analysis

Fig. 10 illustrates the evolution of episode reward, cost, actor loss and critic loss during WCSAC training. In each subfigure, the light - colored area represents the range between the maximum and minimum values of the corresponding metric across multiple training processes, which are determined by different random seeds. Meanwhile, the dark - colored line denotes the average value of that metric. As shown in Fig. 10(a), the episode reward experiences a rapid increase in the early training phase, followed by convergence and stabilization around a high reward value ($>=5000$). This indicates that the agent quickly learns an effective control policy for target tracking under constraints and begins to consistently achieve high returns. Furthermore, Fig. 10(b) presents the episode cost curves, which reflect the number of specific constraint violations as shown in Eq. 31 (e.g., forbidden zone violations, actuator limit breaches). Initially, the cost values are high and fluctuate significantly, reflecting frequent violations of safety constraints during early attitude control exploration. However, as training progresses, the episode cost markedly decreases, approaching zero in many episodes. Although occasional spikes in cost still occur after convergence, these are sparse and typically correspond to rare constraint violations due to the balancing operation between effective mission execution and safety operation. In Fig. 10(c), the average actor loss drops sharply at first and then maintains stability, which implies the actor network gradually optimizes its policy. In Fig. 10(d), the average critic loss also shows a decreasing trend and becomes stable eventually, meaning the critic network accurately evaluates the rewards of actions to support policy learning.

Fig. 11 illustrates the desired quaternion for tracking calculated by Eq. 6. We adopt multiple quantitative metrics to evaluate both mission reward and safety compliance, as well as the tracking performance of the comparison algorithms:

- **Reward**: The accumulated reward over one testing trajectory.
- **Cost**: The accumulated cost over one testing trajectory, including the cost of violating attitude constraint and angular velocity constraint.
- **Reduction**: Percentage reduction in constraint violations compared to baseline method (SAC).
- **Tracking Time**: The duration of time steps during which the target remains within the field-of-view of the satellite payload.

As shown in Table 5, we recorded all metrics in five testing cases with different initial conditions and actor faults, as well as the average performance across all cases. SAC achieves a relatively high average reward, but it incurs a large cost due to the omission of safety constraints, indicating poor safety compliance. SACLag slightly reduces the cost by introducing a Lagrangian-based optimization style, but at the expense of lower tracking reward. SACPID and TD3PID also suffer from degraded



(a) Angle (vanilla SAC in Case1) (b) Angle (WCSAC in Case1) (c) Tracking Time (Case1)

(d) Angle (vanilla SAC in Case2) (e) Angle (WCSAC in Case2) (f) Tracking Time (Case2)

(g) Angle (vanilla SAC in Case3) (h) Angle (WCSAC in Case3) (i) Tracking Time (Case3)

(j) Angle (vanilla SAC in Case4) (k) Angle (WCSAC in Case4) (l) Tracking Time (Case4)

(m) Angle (vanilla SAC in Case5) (n) Angle (WCSAC in Case5) (o) Tracking Time (Case5)

Fig. 17. Tracking performance illustration across five testing cases.

Table 5
Performance of Comparison Algorithms.

Algorithm	Case 1				Case 2				Case 3				Case 4				Case 5				Average			
	Reward	Cost	Reduction	Time	Reward	Cost	Reduction	Time	Reward	Cost	Reduction	Time	Reward	Cost	Reduction	Time	Reward	Cost	Reduction	Time	Reward	Cost	Reduction	Time
SAC	5256.554	78	–	526	5222.960	92	–	524	5187.667	51	–	520	5206.574	67	–	521	5251.204	59	–	526	5224.991	69.4	–	523.4
SACLag	5228.547	14	82.05%	526	5080.253	0	100%	512	4438.016	0	100%	439	5102.367	11	83.58%	512	5249.147	5	91.52%	528	5019.665	6	91.42%	503.4
SACPID	5258.937	14	82.05%	526	4134.535	0	100%	404	5187.768	20	60.78%	519	5200.195	3	95.52%	520	5155.973	8	86.44%	515	4987.481	9	81.95%	496.8
TD3PID	4762.332	69	11.53%	475	2449.025	0	100%	226	1178.093	3	94.11%	73	4585.918	66	1.49%	458	1370.667	83	–	109	2869.207	44.2	–	268.2
Reward Penalty	–	13	83.33%	527	–	0	100%	525	–	0	100%	524	–	0	100%	521	–	0	100%	522	–	2.6	%	523.8
WCSAC	5242.276	2	97.43%	526	5239.958	0	100%	527	5252.114	0	100%	527	5187.197	0	100%	521	5148.286	0	100%	518	5213.966	0.4	91.48%	523.8

Table 6
Fault Identification Performance of Comparison Methods.

Fault Identification Method	Estimation Error		Inference Time (s)
	MSE	MAE	
Least Squares Filter	4.81×10^{-4}	1.26×10^{-2}	4.42×10^{-4}
Gradient Descend	3.98×10^{-4}	1.09×10^{-2}	7.60×10^{-1}
Neural Network in [32]	2.35×10^{-4}	7.39×10^{-3}	3.98×10^{-3}
Neural Network in Fig. 6 (Ours)	1.92×10^{-4}	6.43×10^{-3}	7.49×10^{-3}

reward and unstable cost performance. The reward-penalty approach achieves a relatively low average cost and high tracking time, but the cost is still high in some cases [3] In contrast, the proposed WCSAC framework achieves the best trade-off: it maintains a high average reward while keeping the lowest average cost, and a stable tracking duration across different cases. This demonstrates that WCSAC effectively balances safety and tracking performance under actuator faults, enabling fault-tolerant attitude tracking with minimal manual intervention.

In addition, Figs. 12, 13, 14, 15, 16 and 17 present the dynamics responses and tracking performance of the WCSAC-based agile satellite agent under five representative testing cases, respectively. The curves include actual torque (influenced by actuator faults), the angular velocity, and the current quaternion and error quaternion of the imaging satellite. In all cases, the actual torque output and angular velocity exhibit sharp transients at the beginning of the episode, reflecting the satellite's rapid attitude maneuvering toward the desired orientation. Once the angle falls below the payload's half angle ($angle < \theta_{tracking}$ as shown in Fig. 17), the target enters the sensor's field of view, and the satellite transitions into a fine-tuning phase. During this stage, both torque output and angular velocity gradually stabilize. Minor fluctuations are observed intermittently, corresponding to the agent's maneuvering actions to avoid attitude constraints. Moreover, the convergence behavior of the error quaternion supports this interpretation. Due to the reward function being designed to encourage tracking rather than strict convergence to the specific quaternion $[0, 0, 0, 1]^T$ like traditional attitude control problem [21,50], a small residual tracking error remains, which aligns with the behavior shown in the plots. Furthermore, as shown in Fig. 17, we also compare the tracking performance of vanilla SAC and WCSAC across five cases, focusing on the angle between the payload sightline and the target, as well as the angles to multiple forbidden zones. Taking Case 2 (subfig d,e) as an example, vanilla SAC frequently violates attitude constraints due to its lack of constraint awareness. In contrast, WCSAC effectively maintains all forbidden zones outside the forbidden half-angle threshold ($angle > \theta_{forbidden}$) while keeping the target within the payload half-angle limit ($angle < \theta_{tracking}$). This ensures safe tracking and allows the cumulative tracking time to increase steadily throughout the episode, except during the initial large-angle maneuver and brief constraint-avoidance events.

The performance illustrations indicate that the agent successfully maintains continuous tracking performance on the space target across diverse fault scenarios within LEO operational envelope, demonstrating both agility and fault tolerance.

5.3. Ablation experiment

In this subsection, we conduct ablation experiments to illustrate the effectiveness and efficiency of the proposed neural network-based fault identification method. Specifically, the satellite agent's initial attitude, angular velocity, and actuator faults are randomized to simulate diverse fault conditions. Three fault identification methods–Least Squares Filter, Gradient Descent, and the proposed Neural Network–are evaluated in terms of estimation accuracy (measured by MSE and MAE) and

[3] We don't evaluate the reward term of reward penalty due to the correction in Eq. 38.

Table 7
Performance of Different Faults Estimation.

Different Estimation	Case 1			Case 2			Case 3			Case 4			Case 5			Average		
	Reward	Cost	Time	Reward	Cost	Time	Reward	Cost	Time	Reward	Cost	Time	Reward	Cost	Time	Reward	Cost	Time
a	5242.276	2	526	5239.958	0	527	5252.114	0	527	5187.197	0	521	5148.286	0	518	5213.966	0.4	523.8
b	5237.362	2	525	5169.650	74	522	5194.318	76	521	4759.788	0	473	4772.785	27	480	5026.780	35.8	504.2
c	1366.545	222	95	5170.510	241	520	2052.011	320	173	1948.189	0	160	5095.170	31	512	3126.485	162.8	292.0
d	3641.631	5	352	5214.555	13	526	1760.128	347	125	4277.481	0	423	5101.095	4	513	3998.978	73.8	387.8
e	5242.987	2	526	5240.173	0	527	5251.913	0	527	5196.789	0	522	5147.666	0	518	5215.905	0.4	524.0

computational efficiency (measured by inference time). This experiment is repeated over 5000 runs, and the averaged results are reported in Table 6. As shown in the table, our proposed neural network architecture achieves the lowest estimation error and relatively short inference time, demonstrating both its accuracy and real-time suitability compared with state-of-the-art FDI methods.

In addition, we further evaluate the impact of fault identification quality on the downstream reinforcement learning-based control in a satellite tracking scenario. We consider five estimation configurations: [a] neural network estimated faults ($\hat{e}_l, \hat{u}_a$), [b] none effectiveness loss and random additive bias, [c] random effectiveness loss and non additive bias, [d] random effectiveness loss and random additive bias, and [e] truth fault faults (e_l, u_a). As reported in Table 7, the neural network-based estimation [a] yields a comparable performance to the ideal condition [e] in terms of tracking reward, cost, and execution time. In contrast, the configurations [b]–[d] relying on random or incomplete fault assumptions lead to significant degradation in control performance, especially in terms of reward drop and increased maneuvering cost. This result confirms that accurate and fast fault identification not only improves state estimation but also significantly benefits the decision-making and task execution performance of reinforcement learning-based control policies.

6. Conclusion

This paper presents a fault-tolerant attitude control framework for an agile satellite performing persistent tracking of non-cooperative space targets under multiple operational constraints and potential actuator faults. By decomposing the tracking mission into three distinct stages and formulating the control task as a Constrained Markov Decision Process, we systematically address complex real-world challenges, including attitude forbidden zones, angular velocity limits, torque saturation, and actuator faults. A data-driven fault identification module is integrated to detect both effectiveness degradation and additive actuator faults in real time, while a safe reinforcement learning algorithm is employed to learn robust and adaptive control policies that ensure safe, continuous tracking. Extensive simulation results demonstrate that the proposed method enables agile satellites to achieve accurate and stable tracking performance while satisfying all critical constraints, even in the presence of actuator anomalies. This work provides a promising foundation for the deployment of intelligent, fault-tolerant tracking systems in future autonomous space missions.

Regarding the limitations of this study, both fault identification and tracking experiments were conducted using simulation programs rather than onboard hardware. Furthermore, the neural network requires pretrained prior knowledge, making it highly sensitive to system parameter uncertainties. In future work, we will explore a reliable FDI algorithm that does not require prior knowledge and test it on reaction wheels in the laboratory to empower intelligent satellite applications. In addition, we plan to extend the framework to more complex multi-target tracking scenarios and address generalization across varying forbidden zone counts through network architectural modifications. Moreover, incorporating noise-aware decision-making and robust onboard autonomy will further enhance the system's resilience in noisy space environments.

CRediT authorship contribution statement

Wenlong Lu: Methodology, Investigation; **Hongji Zhuang:** Methodology, Investigation; **Ziyao Geng:** Methodology, Investigation; **Qiang Shen:** Methodology, Investigation; **Zhongcheng Mu:** Resources; **Shufan Wu:** Supervision, Funding acquisition; **Vladimir Y. Razoumny:** Funding acquisition; **Yury N. Razoumny:** Funding acquisition.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] B. Wang, S. Li, J. Mu, X. Hao, W. Zhu, J. Hu, Research advancements in key technologies for space-based situational awareness, Space: Sci. Technol. (2022).
[2] L.U. Wenlong, G. Ziyao, M.U. Zhongcheng, W.U. Shufan, N.R. Yury, et al., Optimizing spacecraft collision avoidance maneuvers under risk uncertainty from multiple space debris, Chin. J. Aeronaut. (2025) 103814.
[3] C. Wang, L. Jiang, M. Li, X. Ren, Z. Wang, Slow-Spinning spacecraft cross-Range scaling and attitude estimation based on sequential ISAR images, IEEE Trans. Aerosp. Electron. Syst. 59 (6) (2023) 7469–7485. https://doi.org/10.1109/TAES.2023.3291337
[4] H. Tang, C. Liu, J. Liu, P. Zhang, W. Hu, Real-time monocular 3D pose tracking for non-cooperative spacecraft in close range, IEEE Trans. Instrum. Meas. (2025) 1. https://doi.org/10.1109/TIM.2025.3568937
[5] X. Cao, N. Li, S. Qiu, C. Li, Research on the method of searching and tracking of the time-sensitive target through the mega-constellation, Aerosp. Sci. Technol. 137 (2023) 108299.
[6] E. Altman, Constrained Markov decision processes, Routledge, 2021.
[7] Q. Yang, T.D. Simão, S.H. Tindemans, M.T.J. Spaan, WCSAC: Worst-case soft actor critic for safety-constrained reinforcement learning, in: Proceedings of the AAAI Conference on Artificial Intelligence, 35, 2021, pp. 10639–10646.
[8] D. Zhou, G. Sun, W. Lei, L. Wu, Space noncooperative object active tracking with deep reinforcement learning, IEEE Trans. Aerosp. Electron. Syst. 58 (6) (2022) 4902–4916.
[9] Z. Yu, X. Su, G. Sun, Fractional order meta-Reinforcement learning for space non-cooperative object active visual tracking, IEEE Trans. Aerosp. Electron. Syst. (2024).
[10] Y.-H. Wu, F. Han, M.-H. Zheng, F. Wang, B. Hua, Z.-M. Chen, Y.-H. Cheng, Attitude tracking control for a space moving target with high dynamic performance using hybrid actuator, Aerosp. Sci. Technol. 78 (2018) 102–117.
[11] W.U. Yunhua, M. Zheng, H.E. Wei, W. Feng, C. Zhiming, H. Bing, High precision attitude dynamic tracking control of a moving space target, Chin. J. Aeronaut. 32 (10) (2019) 2324–2336.
[12] M. Zheng, Y. Wu, C. Li, Reinforcement learning strategy for spacecraft attitude hyperagile tracking control with uncertainties, Aerosp. Sci. Technol. 119 (2021) 107126.
[13] Z. Lin, B. Wu, D. Wang, Specific tracking control of rotating target spacecraft under safe motion constraints, IEEE Trans. Aerosp. Electron. Syst. 59 (3) (2023) 2422–2438. https://doi.org/10.1109/TAES.2022.3214799
[14] W. Lu, W. Gao, B. Liu, W. Niu, D. Wang, Y. Li, X. Peng, Z. Yang, Reinforcement learning driven time-sensitive moving target tracking of intelligent agile satellite, IEEE Trans. Aerosp. Electron. Syst. (2024).
[15] W. Lu, W. Gao, B. Liu, W. Niu, X. Peng, Z. Yang, Y. Song, Parallel dual adaptive genetic algorithm: a method for satellite constellation task assignment in time-sensitive target tracking, Adv. Space Res. 74 (10) (2024) 5192–5213.
[16] Y. Tian, Q. Hu, X. Shao, Adaptive fault-tolerant control for attitude reorientation under complex attitude constraints, Aerosp. Sci. Technol. 121 (2022) 107332.
[17] Q. Hu, Y. Liu, H. Dong, Y. Zhang, Saturated attitude control for rigid spacecraft under attitude constraints, J. Guid., Control, Dyn. 43 (4) (2020) 790–805.
[18] Z. Kang, Q. Shen, S. Wu, C.J. Damaren, Saturated attitude control of multispacecraft systems on SO (3) subject to mixed attitude constraints with arbitrary initial attitude, IEEE Trans. Aerosp. Electron. Syst. 59 (5) (2023) 5158–5173.
[19] S. Wang, Q. Shen, H. Li, R. Niu, Finite-Horizon active fault isolation and identification for spacecraft attitude control systems with multiple constraints, IEEE Trans. Aerosp. Electron. Syst. (2025).
[20] C.R. Mclnnes, Large angle slew maneuvers with autonomous sun vector avoidance, J. Guid., Control, Dyn. 17 (4) (1994) 875–877.
[21] Q. Shen, C. Yue, C.H. Goh, D. Wang, Active fault-tolerant control system design for spacecraft attitude maneuvers with actuator saturation and faults, IEEE Trans. Ind. Electron. 66 (5) (2018) 3763–3772.
[22] J. Elkins, R. Sood, C. Rumpf, Adaptive continuous control of spacecraft attitude using deep reinforcement learning, in: 2020 AAS/AIAA Astrodynamics Specialist Conference, AIAA Reston, VA, 2020, pp. 420–475.
[23] J. Elkins, R. Sood, C. Rumpf, Autonomous spacecraft attitude control using deep reinforcement learning, in: 71St International Astronautical Congress (IAC), 2020, 2020.
[24] W. Lu, B. Liu, Z. Mu, S. Wu, Y. Song, V.Y. Razoumny, Multi-Satellite scheduling for stereo tracking of moving targets via parallel island differential evolutionary algorithm, IEEE Trans. Aerosp. Electron. Syst. (2025).
[25] H. Zhuang, W. Lu, Q. Shen, S. Wu, V.Y. Razoumny, Y.N. Razoumny, Off-policy reinforcement learning control for space manipulators based on object detection via convolutional neural networks, Aerosp. Sci. Technol. (2025) 110914.
[26] H. Zhuang, W. Lu, Q. Shen, S. Wu, V.Y. Razoumny, Y.N. Razoumny, Heterogeneous multi-space manipulator cooperative control in task space via off-policy reinforcement learning, Acta Astronaut. (2025).
[27] H. Zhuang, J. Hou, Q. Shen, S. Wu, V.Y. Razoumny, Y.N. Razoumny, Event-Triggered image-Space tracking control of space manipulators using off-Policy reinforcement learning with disturbance observers, IEEE Trans. Aerosp. Electron. Syst. (2025).
[28] Q. Shen, D. Wang, S. Zhu, K. Poh, Finite-time fault-tolerant attitude stabilization for spacecraft with actuator saturation, IEEE Trans. Aerosp. Electron. Syst. 51 (3) (2015) 2390–2405.
[29] C. Yue, T. Huo, M. Lu, Q. Shen, C. Li, X. Chen, X. Cao, A systematic method for constrained attitude control under input saturation, IEEE Trans. Aerosp. Electron. Syst. 59 (5) (2023) 6005–6015.
[30] Z. Ma, Y. Wang, Y. Yang, Z. Wang, L. Tang, S. Ackland, Reinforcement learning-based satellite attitude stabilization method for non-cooperative target capturing, Sensors 18 (12) (2018) 4331.
[31] Y. Liu, G. Ma, Y. Lyu, P. Wang, Neural network-based reinforcement learning control for combined spacecraft attitude tracking maneuvers, Neurocomputing 484 (2022) 67–78.
[32] W. Lu, Z. Geng, H. Zhuang, Q. Shen, S. Wu, V.Y. Razoumny, Y.N. Razoumny, Neural network-Based fault identification and reinforcement learning for spacecraft fault-Tolerant control, IEEE Trans. Instrum. Meas. (2025) 1. https://doi.org/10.1109/TIM.2025.3597688
[33] U.J. Ravaioli, J. Cunningham, J. McCarroll, V. Gangal, K. Dunlap, K.L. Hobbs, Safe reinforcement learning benchmark environments for aerospace control systems, in: 2022 IEEE Aerospace Conference (AERO), 2022, pp. 1–20. https://doi.org/10.1109/AERO53065.2022.9843750
[34] C. Mu, S. Liu, M. Lu, Z. Liu, L. Cui, K. Wang, Autonomous spacecraft collision avoidance with a variable number of space debris based on safe reinforcement learning, Aerosp. Sci. Technol. 149 (2024) 109131.
[35] K.P. Sharma, I. Kumar, P.P. Singh, K. Anbazhagan, H.M. Albarakati, M.W. Bhatt, A.A. Ziyadullayevich, A. Rana, et al., Advancing spacecraft rendezvous and docking through safety reinforcement learning and ubiquitous learning principles, Comput. Human. Behav. 153 (2024) 108110.
[36] C. Han, Y. Zhang, S. Bai, X. Sun, X. Wang, Novel method to calculate satellite visibility for an arbitrary sensor field, Aerosp. Sci. Technol. 112 (2021) 106668.
[37] Y. Lian, Y. Gao, G. Zeng, Staring imaging attitude control of small satellites, J. Guid., Control, Dyn. 40 (5) (2017) 1278–1285.
[38] X. Sun, Q. Shen, S. Wu, Partial state feedback MRAC-Based reconfigurable fault-Tolerant control of drag-Free satellite with bounded estimation error, IEEE Trans. Aerosp. Electron. Syst. 59 (5) (2023) 6570–6586.
[39] X. Sun, Q. Shen, S. Wu, Fuzzy supervised learning-Based model-Free adaptive fault-Tolerant spacecraft attitude control with deferred asymmetric constraints, IEEE Trans. Aerosp. Electron. Syst. 59 (6) (2023) 8884–8900.
[40] X. Zhang, H. Zhang, H. Zhou, C. Huang, D. Zhang, C. Ye, J. Zhao, Safe reinforcement learning with dead-ends avoidance and recovery, IEEE Rob. Autom. Lett. 9 (1) (2023) 491–498.
[41] H. Zhang, G. Solak, G.J.G. Lahr, A. Ajoudani, Srl-vic: a variable stiffness-based safe reinforcement learning for contact-rich robotic tasks, IEEE Rob. Autom. Lett. (2024).
[42] Y.C. Tang, J. Zhang, R. Salakhutdinov, Worst cases policy gradients, arXiv:1911.03618 (2019).
[43] T. Haarnoja, A. Zhou, P. Abbeel, S. Levine, Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor, in: International Conference on Machine Learning, PMLR, 2018, pp. 1861–1870.
[44] T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, S. Levine, Soft Actor-Critic Algorithms and Applications, 2019. arXiv:1812.05905
[45] A. Ray, J. Achiam, D. Amodei, Benchmarking safe exploration in deep reinforcement learning, arXiv:1910.01708 7 (1) (2019) 2.
[46] A. Stooke, J. Achiam, P. Abbeel, Responsive safety in reinforcement learning by pid lagrangian methods, in: International Conference on Machine Learning, PMLR, 2020, pp. 9133–9143.
[47] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al., Pytorch: an imperative style, high-performance deep learning library, Adv. Neural Inf. Process. Syst. 32 (2019).
[48] J. Ji, J. Zhou, B. Zhang, J. Dai, X. Pan, R. Sun, W. Huang, Y. Geng, M. Liu, Y. Yang, Omnisafe: an infrastructure for accelerating safe reinforcement learning research, J. Mach. Learn. Res. 25 (285) (2024) 1–6.
[49] W. Chu, S. Wu, Z. Wu, Y. Wang, Least square based ensemble deep learning for inertia tensor identification of combined spacecraft, Aerosp. Sci. Technol. 106 (2020) 106189.
[50] S. Zhu, D. Wang, Q. Shen, E.K. Poh, Satellite attitude stabilization control with actuator faults, J. Guid., Control, Dyn. 40 (5) (2017) 1304–1313.